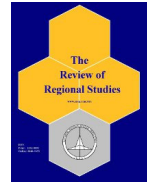




The Review of Regional Studies

The Official Journal of the Southern Regional Science Association



Regional Economic Output and Public Surface Transportation Infrastructure: A Spatial Granger Approach*

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Abstract: This paper expands upon the classical Granger causality test with spatial dimensions to investigate the causal linkages between regional economic output and surface transportation infrastructure with a focus on the Northeast Corridor in the United States. To improve the validity of estimation, a panel data set of surface transportation infrastructure including highways, public intercity railways, and public transit, which are measured in real monetary value for the period 1991–2009, is adopted. Granger causality tests with integrations of various forms of spatial spillover interactions are analyzed in different spatial vector autoregressive models. The empirical results reveal that the spatial Granger causality test is able to capture a broader geographic scale of causality than the classical approach, but the causality between regional economic output and surface transportation infrastructure is sensitive to the model specifications.

Keywords: regional economic output, public surface transportation, Granger causality, spillovers

JEL Codes: C23, R11, R42, R53

1. INTRODUCTION

Understanding the linkages between regional economic performance and surface transportation infrastructure has long been an important task for policymakers and academic scholars. A valid knowledge of such a linkage may help decision makers to allocate funding resources wisely and effectively among different modes of public infrastructure so that greater economic benefits can be achieved. Over the past decades, the literature regarding the empirical assessments of public infrastructure's economic benefits have expanded exponentially since the milestone studies by Aschauer (1989, 1990) and Munnell (1992), who confirmed that public infrastructure has a significant positive impact on economic output. Still, some recent studies have found that the actual impact is much smaller and in some cases is even insignificant (Gramlich, 1994; Gramlich, 2001; Harmatuck, 1996; Nadiri and Mamuneas, 1996; Fernald, 1999; Boarnet, 1997; Boarnet and Haughwout, 2000).

* We are grateful to editors Michael L. Lahr and Gianfranco Piras and to three anonymous referees for their invaluable comments. We also highly appreciate technical support from Donald Lacombe and Rajendra Kulkarni. An earlier version of this paper was presented at the VII World Conference of the Spatial Econometrics Association in 2013. Any error or interpretation mistake is the authors' own responsibility.

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